■ C++ Complete Roadmap — Beginner to Placement Level (Compact Version)

■ PHASE 1: Foundation Booster (Week 1–2)

■ Goal: Learn syntax, basics, and how C++ works.

■ Topics:

[ ] Basic structure of a C++ program
[ ] Input/output (cin, cout, endl)
[ ] Data types and variables
[ ] Operators (arithmetic, logical, relational, bitwise)
[ ] Conditional statements (if, else, switch)
[ ] Loops (for, while, do-while)
[ ] Functions — declaration, definition, and calling

■ Tasks / Practice:

[ ] Calculator program
[ ] Grade calculator
[ ] Number guessing game

■■ PHASE 2: Core Concepts (Week 3–4)

■ Goal: Understand memory, scope, and data flow.

■ Topics:

[ ] Arrays (1D, 2D)
[ ] Strings (char arrays vs string class)
[ ] Pointers and references
[ ] Memory allocation (new, delete)
[ ] Structures (struct)
[ ] Function overloading

■ Tasks / Practice:

[ ] Reverse an array
[ ] Palindrome checker
[ ] Pointer-based swap function

■ PHASE 3: Object-Oriented Programming (Week 5–7)

■ Goal: Write modular, reusable, and maintainable C++ code.

■ Topics:

[ ] Class and objects
[ ] Constructors & destructors
[ ] Access specifiers (private, public, protected)
[ ] Static members and this pointer
[ ] Inheritance (single, multiple, multilevel, hybrid)
[ ] Polymorphism and virtual functions
[ ] Encapsulation and abstraction

■ Tasks / Practice:

[ ] Bank management system
[ ] Student management system
[ ] Shape area calculator

■ PHASE 4: Advanced Concepts (Week 8–9)

■ Goal: Understand modern and deeper C++ concepts.

■ Topics:

[ ] Operator overloading
[ ] Friend functions and classes
[ ] Templates (function & class templates)
[ ] Exception handling (try, catch, throw)
[ ] File handling (ifstream, ofstream, fstream)
[ ] Namespace and scope resolution
[ ] Inline functions, macros, typedef, enum

■ Tasks / Practice:

[ ] File-based student database
[ ] Generic calculator using templates

■ PHASE 5: Standard Template Library (Week 10–12)

■ Goal: Master STL — your strongest weapon in interviews.

■ Topics:

[ ] STL Containers: vector, list, deque, stack, queue, priority_queue
[ ] Associative Containers: set, unordered_set, map, unordered_map
[ ] Iterators
[ ] Algorithms (sort, find, reverse, binary_search, etc.)
[ ] Pairs, tuples, lambdas

■ Tasks / Practice:

[ ] Sort students by marks using vector
[ ] Frequency counter using map
[ ] Bracket balance checker using stack

■ PHASE 6: DSA using C++ (Week 13+)

■ Goal: Crack coding rounds confidently.

■ Topics:

[ ] Arrays and Strings
[ ] Linked List (single, double, circular)
[ ] Stack and Queue
[ ] Trees and Binary Search Trees
[ ] Heaps and Hashing
[ ] Graphs (BFS, DFS, shortest path basics)
[ ] Sorting and Searching algorithms
[ ] Recursion, Backtracking, Greedy, DP

■ Tasks / Practice:

[ ] Practice on LeetCode
[ ] Solve GFG topic-wise problems
[ ] Attempt CodeStudio challenges

■ PHASE 7: Real-world Applications

■ Goal: Apply C++ in real projects.

■ Topics:

[ ] CLI-based Expense Tracker
[ ] File-based Todo App
[ ] Simple Database using OOP + File handling
[ ] Mini console games (Snake, Tic Tac Toe)

■ Tasks / Practice:

[ ] Build one mini project combining OOP + File Handling

■ RESOURCES & REFERENCES

- YouTube: CodeWithHarry, Apna College, Striver
- Books: Let Us C++, C++ Primer, Effective C++
- Practice Sites: LeetCode, GFG, CodeStudio

■ Track your progress by checking each [ ] box once you complete a topic.